

Microbes Affecting the Musculoskeletal and Integumentary Systems

I. Microbial Diseases of Skin, Soft Tissues, Muscles and Wounds

○ Introduction

○ Bacterial Diseases

- Folliculitis
- Staphylococcal Scalded Skin Syndrome
- Impetigo (Pyoderma) and Erysipelas
- Cellulitis
- Acne
- Cat Scratch Disease
- Pseudomonas Infection
- Cutaneous Anthrax
- Necrotizing Fasciitis
- Gas gangrene and myonecrosis
- Leprosy (reading assignment)
- 1^o and 2^o stages of syphilis (reading assignment)

Microbial Diseases of Skin ...

○ Viral Diseases

- Herpes Infections
- Warts (papillomas)
- Chickenpox (varicella) and Shingles (zoster)
- Rubella
- Measles (Rubeola)
- Other Viral Rashes
 - Erythema Infectiosum
 - Roseola
 - Coxsackievirus Infection
 - FHM disease

Microbial Diseases of Skin ...

○ Mycoses of the Skin, Soft tissues, Hair and Nails

- Superficial Mycoses
 - Pityriasis versicolor
 - Piedra
 - Tinea nigra palmaris
- Cutaneous Mycoses (Dermatophytoses)
 - Tinea capitis
 - Tinea corporis
 - Tinea pedis
 - Tinea cruris
 - Tinea unguium (onychomycosis)
- Subcutaneous mycoses
 - Sporothricosis
 - Mycetoma
 - Chromoblastomycosis
 - Phaeohyphomycosis
- Cutaneous and mucocutaneous candidiasis

Microbial Diseases of Skin ...

○ Parasitic Infestations of the Skin, soft tissues and muscles

- Leishmaniasis
- Onchocerciasis
- Cutaneous larval migrans
- Trichinella spiralis infection
- Scabies
- Myiasis

II. Microbial Diseases of Bones and Joints

- Osteomyelitis
- Infective/septic arthritis

Introduction

- In addition to being a structural barrier, the skin is colonized by an array of organisms which forms its **normal flora**.
 - the relatively arid areas of the forearm and back are colonized with fewer organisms, predominantly aerobic Gram-positive bacteria (*Staphylococcus*, *Micrococcus*, *Diphtheroids*) and yeasts (*Malassezia*).
 - in the moister areas, such as the groin and the armpit, the organisms are more numerous and more varied and include Gram-negative bacteria.
- The normal flora of the skin play an important role in defending the surface from foreign invaders.
 - compete with potential pathogens for nutrients and space and produce enzymes and chemicals that interfere with the growth of other microbes

Introduction ...

- Microbial disease of the skin may result from any of three lines of attack:
 - breach of intact skin, allowing infection from the outside
 - skin manifestations of systemic infections, which may arise:
 - as a result of blood-borne spread from the infected focus to the skin
 - by direct extension, e.g.,
 - draining sinuses from actinomycotic lesions
 - necrotizing anaerobic infection from intra-abdominal sepsis
 - toxin-mediated skin damage due to production of a microbial toxin at another site in the body (e.g., scarlet fever, toxic shock syndrome)

Introduction ...

- Breaches in the skin range from microscopic to major trauma, which may be:
 - accidental (e.g., lacerations or burns)
 - intentional (e.g., surgery).
 - Hospitalized patients are liable to other skin breaches
pressure sores intravenous catheter insertions

Bacterial Diseases

❑ Folliculitis

- an infection of a hair follicle in which the base of the follicle becomes red, swollen, and pus filled (**pimple**).
- when it occurs at the base of an eyelid, it is called a **sty**.
- a **furuncle** or **boil**, is a large, painful, raised nodular extension of folliculitis resulting from spread of the infection into surrounding tissues.
- when several furuncles join together—more frequently in areas where the skin is thick, such as at the back of the neck—they form a **carbuncle**

Folliculitis ...



Folliculitis. Note small yellow pustules around hair follicles, with little surrounding inflammation



Furuncle. Note larger collection of yellow pus, raised from skin surface with definite red inflamed edge



S. aureus carbuncle

Folliculitis ...

○Pathogen and Virulence Factors

- *Staphylococcus* is the most common cause of folliculitis and associated infections of the skin
 - facultatively anaerobic, Gram-positive bacteria whose spherical cells are typically clustered in grapelike arrangements
 - salt tolerant: capable of growing in 10% NaCl
 - tolerant of drying out, solar radiation, and heat (up to 60°C for 30 min)
- Two species of *Staphylococcus* are typically found on human skin as well as in the upper respiratory, gastrointestinal, urinary, and genital tracts.
 - *Staphylococcus epidermidis* is a major member of the microbiota, accounting for up to 90% of bacteria on the skin.
 - The more virulent *Staphylococcus aureus*, which often grows in nasal passages, produces a variety of disease conditions and symptoms

Folliculitis ...

- *Staphylococci* have at least three categories of virulence factors: enzymes, structural components and toxins
 - **Enzymes**
 - **Coagulase** clots blood, which may hide the bacterium from phagocytes.
 - **Hyaluronidase** breaks down hyaluronic acid, a major component of the matrix between cells, thus enabling the bacterium to spread between cells throughout the body.
 - **Staphylokinase** - dissolves blood clots, which also allows staphylococci to spread to new locations.
 - **Lipases**, which are present in *S. epidermidis* as well, digest lipids, including sebum, providing staphylococci with food on the surface of skin, in hair follicles, and in sebaceous glands.
 - **Beta-Lactamase** – inactivates beta-lactam antibiotics

Folliculitis ...

- **Toxins**

- **cytolytic toxins** disrupt the cytoplasmic membranes of a variety of cells.
- **leukocidin** kills leukocytes, providing Staphylococcus with additional protection against phagocytosis
- **epidermal cell differentiation inhibitor** is a protein that induces large holes in the linings of blood vessels, allowing access for the bacterium to invade body tissues
- **exfoliative toxin** - causes staphylococcal scalded skin syndrome
- **toxic shock syndrome toxin** – staphylococcal toxic shock syndrome

Folliculitis ...

- **Structural Defenses**

- loosely organized polysaccharide **slime layer** or **capsule**
 - inhibit chemotaxis of and phagocytosis
 - facilitates attachment of staphylococcal biofilms to artificial surfaces such as catheters

- **protein A**

- inhibits opsonization
- inhibits the C' cascade, which is triggered by antibody molecules bound to antigen

Comparison of Virulence Factors of Two Staphylococcal Species

Virulence Factor	<i>S. aureus</i>	<i>S. epidermidis</i>
Enzymes		
Coagulase	+	—
Staphylokinase	+	—
Lipase	+	+
β-Lactamase	Present in 90% of strains	—
Factors That Inhibit Phagocytosis		
Polysaccharide slime layer	+	+
Protein A on cell surface	+	—
Toxins		
Cytolytic toxins	+	—
Leukocidin	+	—
Epidermal cell differentiation inhibitor	+	—
Exfoliative toxin	Present in some strains	—
Toxic shock syndrome toxin	Present in some strains	—

Folliculitis ...

○Pathogenesis

- Humans transmit both species of *Staphylococcus* via direct contact between individuals as well as via fomites
- *Staphylococcus* on the surface of the skin grows into hair follicles and invades sebaceous glands which triggers fever and inflammation
 - causes the follicle to enlarge and fill with pus composed of leukocytes, dead cells, and bacteria.
 - infection may spread into the hypodermis to form a furuncle or into neighbouring hair follicles to form a carbuncle.

Folliculitis ...

○Laboratory Diagnosis

- Specimens

- Material scraped from the base of the abscess with a swab or curette (abundant organisms)
- Aspirated pus or superficial specimens collected with swabs contain relatively few organisms

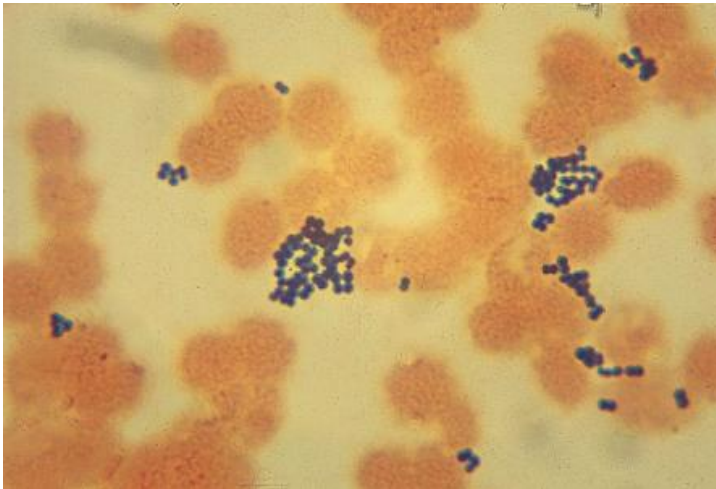
- Microscopy

- *Staphylococci* are **gram-positive cocci** that appear as single cells or small groups of organisms in a clinical specimen

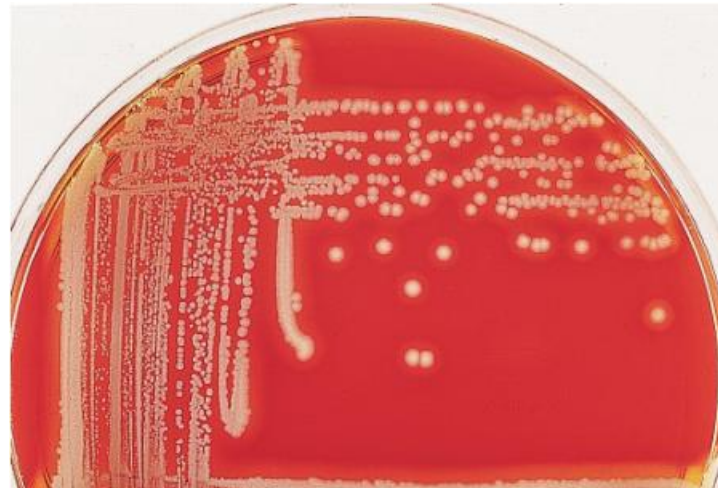
- Culture

- Blood agar plates give typical colonies in 18 hrs. at 37°C
- Media containing 7.5% NaCl inhibit the growth of other bacteria, but not of *S. aureus*

Folliculitis ...



Gram stain of *Staphylococcus aureus*.
Gram- positive cocci in
grape- like clusters.



Staphylococcus aureus on blood agar.
Colonies are cream-colored and opaque
and have a smooth, entire edge. A
zone of beta- hemolysis surrounds the
colony.

Folliculitis ...

- **Catalase Test**

- A drop of H_2O_2 + small amount of bacterial growth on a slide $\Rightarrow$ formation of bubbles (release of O_2) $\Rightarrow$ Positive test

- **Coagulase Test**

- Human/citrated rabbit plasma diluted 1:5 + equal volume of growth from colonies from broth culture $\Rightarrow$ incubated at 37°C $\Rightarrow$ clots (in 1- 4 hrs.) $\Rightarrow$ positive test

- **Serology**

- Antibodies to teichoic acid (chronic infections)

❑ Staphylococcal Scalded Skin Syndrome

- Also called '**Ritter's disease**' in infants and '**Lyell's disease**' or '**toxic epidermal necrolysis**' in older children
- Caused by strains of *S. aureus* producing a toxin known as '**exfoliatin**' or '**scalded skin syndrome toxin**'.
- Cells of the outer epidermis separate from one another and from the underlying tissue
- Involves a reddening and wrinkling of the skin
 - begins near the mouth
 - spreads over the entire body
 - followed by large blisters that contain clear fluid **lacking bacteria or white blood cells**.
 - within two days, the affected outer epidermis peels off in sheets

SSSS ...



SSSS ...

○Pathogen and Virulence Factors

- 5% of strains of *S. aureus* secrete one or two distinct exfoliative toxins that cause SSSS.
- Each toxin causes the **dissolution of epidermal desmosomes**, which are intercellular bridge proteins that hold together cytoplasmic membranes of adjoining cells.
- Both toxins affect keratinized cells of the epidermis

○Pathogenesis

- the blood carries exfoliative toxins from sites of infection throughout the body (toxaemia)
- the body restores the lost epidermis within 7–10 days after protective antibodies circulate in the blood
- scarring does not occur because the dermis is not affected by the toxin

SSSS ...

○Diagnosis

- Made on the basis of the distinctive sloughing of the skin.
- Fluid in the blisters of SSSS does not contain *S. aureus*, since the disease is mediated by toxins released from a site of infection that may be elsewhere in the body

□ Impetigo (Pyoderma) and Erysipelas

- **Impetigo** also called **pyoderma** is a contagious disease characterized by small, flattened, red patches that appear primarily on the face and limbs
 - patches develop into oozing, pus-filled vesicles on a red base
 - vesicles eventually break and form a thick honey-coloured crust
 - attached firmly to the skin
 - cause intense itching
 - numerous vesicles at various stages of development characterize impetigo because bacteria from a vesicle spread to adjacent sites on the skin

Impetigo ...

- When such an infection of the skin spreads into surrounding lymph nodes and triggers pain and inflammation, the condition is called **erysipelas**. **Ecthyma** is an ulcerative form of impetigo that extends deeper into the dermis.
 - manifests as reddening of the skin on the face, arms, or legs.
 - the red area has a distinct margin
 - swollen local lymph nodes, pain, fever, chills, and leucocytosis



Impetigo



Erysipelas

Impetigo ...

○Pathogens and Virulence Factors

- *S. aureus* acting alone causes about 80% of impetigo cases
- about 20% of cases involve *Streptococcus pyogenes* alone or in conjunction with *S. aureus*.
- *S. pyogenes* is also the cause of erysipelas
- *S. pyogenes* (group A Streptococcus) is a Gram-positive coccus that forms chains
- *S. pyogenes* has virulence factors similar in action to those of *S. aureus*:
 - **M protein** is a cytoplasmic membrane component that destabilizes complement and interferes with phagocytosis and lysis of the bacterium.
 - **A hyaluronic acid capsule** serves to “camouflage” the bacterium, hiding it from phagocytes since hyaluronic acid is a natural chemical in the body.
 - **Pyrogenic toxins** are proteins that stimulate macrophages and helper lymphocytes to release cytokines that in turn stimulate fever, a widespread rash, and shock

Impetigo ...

○Pathogenesis

- The bacteria of impetigo and erysipelas occasionally colonize the skin and then invade through scratches, abrasions, cold sores, or other wounds
- Some strains of *S. pyogenes* may spread from a site of impetigo or erysipelas into the blood (bacteraemia) and thence into the kidneys where they cause AGN

Impetigo ...

○Lab. Diagnosis

• Gram stain

- the presence of grapelike clusters of Gram-positive cocci indicate staphylococcal impetigo
- the presence of chains of Gram-positive cocci characterizes streptococcal impetigo and erysipelas

• Culture

- in patients with impetigo, the crusted top of the lesion is raised, and the purulent material and base of the lesion are cultured
- *S. pyogenes*
 - forms characteristic small, opalescent colonies surrounded by a large zone of beta hemolysis on sheep blood agar.
 - Bacitracin susceptibility test (sensitive)
 - L-pyrrolidonyl arylamidase (PYR) test (pink colour in the presence of pyrrolidonyl arylamidase)

Impetigo ...



Gram stain of *S. pyogenes*



S. pyogenes on blood agar



Growth of *S. pyogenes*
is inhibited by 0.04U bacitracin disk (right)

□ Cellulitis

- An acute spreading infection of the skin that involves subcutaneous tissues
- Extends deeper than erysipelas and originates either
 - from superficial skin lesions such as boils or ulcers, or
 - following trauma
- Infection develops within a few hours or days of trauma
 - quickly produces a hot red swollen lesion
 - regional lymph nodes are enlarged
 - and the patient suffers malaise, chills and fever

Cellulitis ...

- Majority of cases are caused by *S. pyogenes* and *S. aureus*
- The pathogen causing cellulitis is isolated in only 25–35% of cases
- Attempts can be made to confirm the clinical diagnosis by culture of:
 - aspirates from the advancing edge of the cellulitis
 - the site of trauma (if present)
 - skin biopsies
 - blood.

❑ Acne

- Characterized by blackheads and pimples
- The most common causes of acne are propionibacteria
 - small, Gram-positive, rod-shaped diphtheroids commonly found on the skin
 - produce propionic acid from fermentation of carbohydrates
 - *Propionibacterium acnes* causes acne in 85% of afflicted adolescents and young adults



Acne ...

○Pathogenesis

- *Propionibacterium* grows on sebum within the sebaceous glands of the skin .
- Excessive oil production triggered by the hormones of adolescence, particularly testosterone in males, stimulates the growth of the bacteria
 - secrete chemicals that attract leukocytes and trigger inflammation.
 - leukocytes phagocytize the bacteria and release chemicals that stimulate local inflammation.
- The combination of dead bacteria and dead and living leukocytes makes up the white pus associated with the pimples of acne .
- A blackhead is formed when a plug of dead and dying bacteria blocks the pore
- In cystic acne (severe form), bacteria form inflamed pustules (cysts) that rupture, triggering the formation of scar tissue.
- Acne typically develops on those areas of the skin where sebaceous glands are most numerous, including the face, scalp, neck, chest, back, upper arms, and shoulders

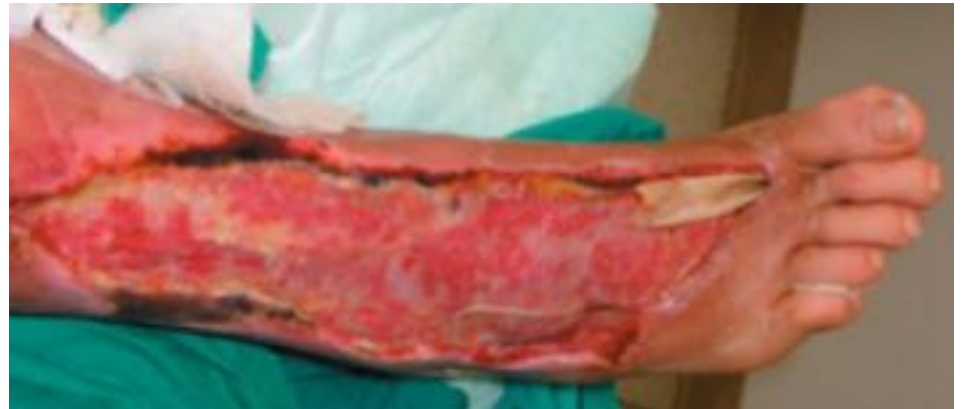
Acne ...

○Diagnosis

- Characteristic pimples and blackheads

❑ Necrotizing Fasciitis

- *S. pyogenes* can cause necrotizing soft tissue infection, sensationalized by the news media as “flesh-eating strep”
- characterized by
 - intense pain and swelling at site of infection
 - discoloration and distension of skin
 - destruction of muscle fascia and fat tissue (necrosis)
 - fever, nausea, malaise, hypotension



Necrotizing Fasciitis ...

○Pathogen and Virulence Factors

- *S. pyogenes* (GAS) is the most common cause
- Strains of *S. pyogenes* that cause necrotizing fasciitis have enzymes and toxins
 - **Deoxyribonucleases** (DNase) which break down DNA
 - **Streptokinases** which dissolve blood clots
 - **Hyaluronidase** which breaks down hyaluronic acid between cells
 - **Exotoxin A** causes toxaemia
 - triggers an overactive immune response and tissue damage
 - **Streptolysin S** is able to kill many kinds of human cell types

Necrotizing Fasciitis ...

○Pathogenesis

- *S. pyogenes* enters the body through wounds and spreads rapidly along muscle fascia
 - secretes the enzymes and toxins
 - destroy tissues at a rate of several centimetres of tissue an hour
 - cause organ failure
 - >50% of patients die
- The usual route of transmission is from person to person

❑ Cat Scratch Disease

- involves fever for a few days, prolonged malaise, and localized swelling at the site of infection and nearby lymph nodes for several months

○ Pathogen, Virulence Factors and Pathogenesis

- *Bartonella henselae*, a Gram-negative aerobic bacillus, causes cat scratch disease.
- Its primary virulence factor is endotoxin (lipid A) found in the outer membrane of Gram-negative bacteria.
- *Bartonella* can grow and reproduce inside red blood cells and in cells lining blood vessel walls.
- Cat scratches or bites, particularly wounds by kittens, introduce the bacterium into the skin. Blood-sucking arthropods such as Fleas may also transmit the bacterium from cats to people.
- In the skin, the bacterium grows intracellularly and releases endotoxin when it dies, which triggers inflammation

Cat Scratch ...

○Diagnosis

- A positive indirect fluorescent antibody test against *Bartonella* antigens confirms a diagnosis of cat scratch disease in individuals who exhibit the characteristic signs and symptoms following exposure to cats.

❑ Pseudomonas Infection

- Common in burn victims
- when *Pseudomonas aeruginosa* invades the bloodstream, it causes
 - fever
 - chills
 - Shock
- Massive infections are often readily diagnosed because the bacterium typically produces a blue-green pigment, pyocyanin, that colors such infections.



Ecthyma gangrenosum
due to a *P. aeruginosa*

Pseudomonas ...

○Pathogen and Virulence Factors

- *Pseudomonas aeruginosa* is a Gram-negative, aerobic bacillus that metabolizes a wide range of organic carbon and nitrogen sources
- almost everywhere: soil, decaying organic matter, swimming pools, hot tubs, sponges, washcloths, and contact lens solutions, sinks, moist foods, vases of cut flowers, sponges, toilets, floor mops, dialysis machines, respirators, and humidifiers
- *P. aeruginosa* has numerous virulence factors

Pseudomonas ...

- *P. aeruginosa* has numerous virulence factors
 - **Fimbriae** and other **Adhesins** enable its attachment to host cells and formation of biofilms.
 - **Neuraminidase** modifies receptors on the host cell in such a way that attachment of fimbriae is enhanced.
 - A **mucoïd polysaccharide capsule** also plays a role in attachment and shields the bacterium from phagocytosis.
 - **Exotoxin A** and **Exoenzyme S**—inhibit eukaryotic protein synthesis, contributing to host cell death.
 - **Elastase** breaks down elastic fibers, degrades complement components, and cleaves IgA and IgG.
 - **Pyocyanin** triggers the formation of superoxide radical ($O_2^{\cdot-}$) and peroxide anion (O_2^{2-})—two reactive forms of oxygen that contribute to tissue damage in *Pseudomonas* infections

Pseudomonas ...

○Pathogenesis

- The surface of a burned area provides a warm, moist environment that is quickly colonized by *P. aeruginosa*
- A thick, scab-like crust naturally forms over the surface of a severe burn.
- *Pseudomonas* growing beneath the crust can move into the blood
 - kills cells and destroys tissues
 - endotoxin mediates fever, vasodilation, inflammation, and shock

Pseudomonas ...

○Laboratory Diagnosis

○Microscopy

- Observation of thin gram-negative rods arranged singly and in pairs is suggestive of *Pseudomonas*

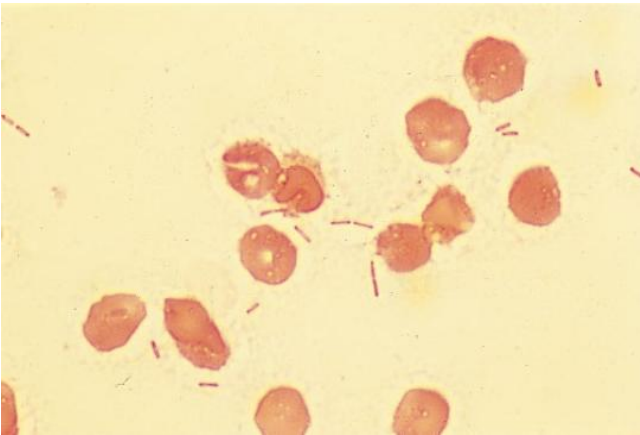
○Culture

- Because *Pseudomonas* has simple nutritional requirements, the bacteria are readily recovered on common isolation media such as blood agar and MacConkey agar

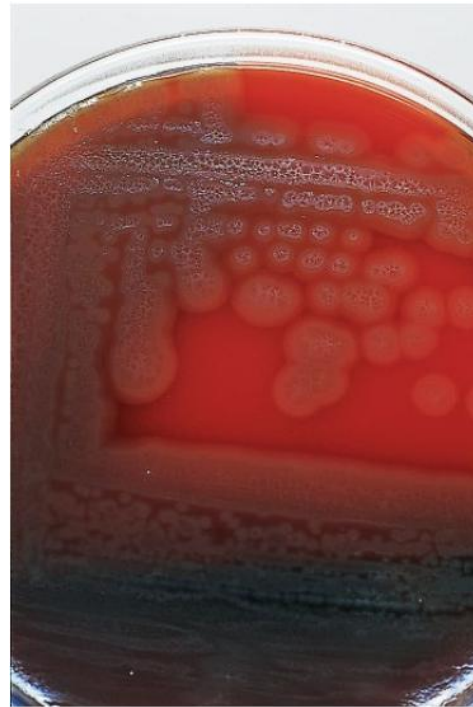
○Identification

- *P. aeruginosa* grows rapidly and has flat colonies with a spreading border
- **β-hemolysis**
- a **blue-green pigmentation** caused by the production of the blue (pyocyanin)
- yellow-green (pyoverdine) pigments
- a characteristic sweet, **grapelike odour**
- positive **oxidase** reaction

Pseudomonas ...



Gram stain of *Pseudomonas aeruginosa*: slender, Gram- negative bacilli with rounded ends



P. aeruginosa on blood agar

❑ Cutaneous Anthrax

- Begins when *Bacillus anthracis* spores, shed by an infected animal, enters a wound in the skin, producing a solid skin nodule
- Cells in the affected area die, and the nodule spreads to form a painless, swollen, black, crusty ulcer called an **eschar**

○Pathogen and Virulence Factors

- *B. anthracis* is a large (1×3 to $8 \mu\text{m}$) aerobic or facultatively anaerobic Gram-positive, spore-forming, encapsulated organism arranged as single or paired rods or as long, serpentine chains.
- although spores are readily observed in 2- to 3-day-old cultures, they are not seen in clinical specimens

Cutaneous Anthrax ...

- the major factors responsible for the virulence of *B. anthracis* are
 - **Capsule** inhibits phagocytosis of replicating cells
 - **Edema toxin** has adenylate cyclase activity and is responsible for the fluid accumulation observed in anthrax
 - **Lethal toxin** has zinc metalloprotease activity and stimulates macrophages to release TNF- α , IL-1 β , and other proinflammatory cytokines

Cutaneous Anthrax ...

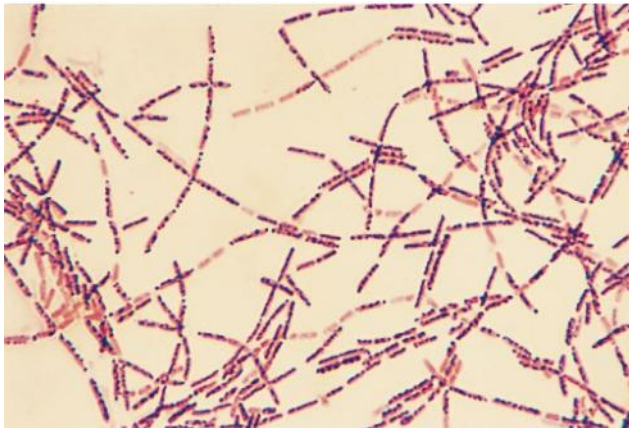


Eschars of cutaneous anthrax

Cutaneous Anthrax ...

- **Laboratory Diagnosis**

- Infections with *B. anthracis* are characterized by overwhelming numbers of organisms present in wounds, involved lymph nodes, and blood.
- Detection of organisms by microscopy and culture is not a problem



Gram stain of *B. anthracis*, showing Gram-positive rods in long chains

❑ Gas Gangrene

- When blood supply to a tissue is interrupted, e.g., by a wound, a condition known as **ischemia** develops.
 - the tissue becomes anaerobic and necrotic
 - if the wound is infected with endospores of anaerobic *Clostridium*, then **gas gangrene** develops

○ Signs and Symptoms

- *Clostridium* produces toxins:
 - kill surrounding tissues
 - providing the bacterium with more nutrients
 - increasing the size of the anaerobic environment necessary for clostridial growth
 - spread into the surrounding tissue, causing the death of muscle and connective tissues

Gas Gangrene ...

- The immediate result is
 - intense pain at the initial site of infection (often in a foot)
 - gas gangrene, which involves blackening of the infected muscle and skin (**myonecrosis**)
 - production of bubbles of hydrogen and carbon dioxide gases that may break out with a frothy brownish fluid
 - the bubbles can be felt under the skin by palpation

Gas Gangrene ...



Gas Gangrene ...

○Pathogen

- *C. perfringens*
 - a large (0.6 to 2.4 × 1.3 to 19.0 μm), rectangular, gram-positive, anaerobic, spore-forming bacillus
 - ubiquitous in soil, water, sewage, and the gastrointestinal tracts of animals and humans
- Colonies of *C. perfringens* are distinctive:
 - rapid, spreading growth on laboratory media
 - β-hemolysis on blood agar

Gas Gangrene ...

○ Virulence Factors

- **Alpha toxin** - a lecithinase (phospholipase C)
 - lyses erythrocytes, platelets, leukocytes, and endothelial cells
 - mediates massive hemolysis, increased vascular permeability and bleeding (augmented by destruction of platelets), tissue destruction, hepatic toxicity, and myocardial dysfunction (bradycardia, hypotension)
- **Beta toxin** is responsible for intestinal stasis, loss of mucosa with formation of necrotic lesions, and progression to necrotizing enteritis
- **Epsilon toxin**, protoxin activated by trypsin and increases the vascular permeability of the GI wall
- **Iota toxin** has necrotic activity and increases vascular permeability.

Gas Gangrene ...

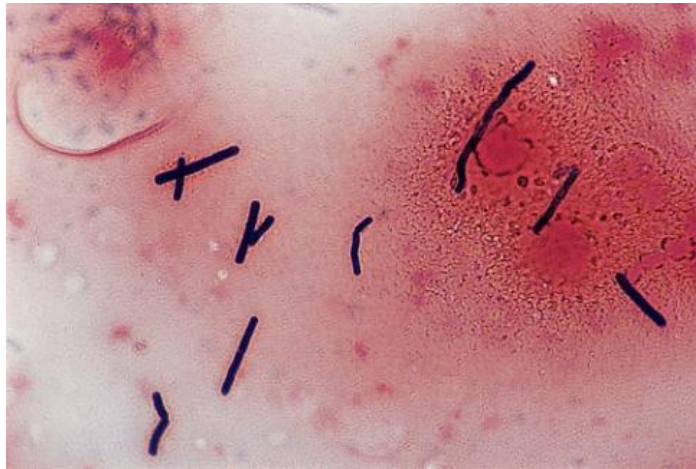
○Pathogenesis

- *Clostridium* does not affect healthy tissues and is not invasive.
- endospores are introduced into a trauma
 - surgical incision, puncture wound, gunshot wound, crushing trauma, abortion, or compound fracture
- endospores germinate in the anaerobic env't and grow on nutrients released by the dead cells.

Gas Gangrene ...

○Diagnosis

- appearance of gas gangrene is usually diagnostic by itself
- detection of large Gram-positive bacilli is confirmatory



Gram stain of *C. perfringens*: large, boxcar- shaped, Gram- positive bacilli; spores are rarely seen

Viral Diseases of the Skin and Wounds

- Exanthems and mucocutaneous lesions caused by viruses can be divided into:
 - those in which the virus remains **restricted to the body surface** at the site of initial infection
 - those in which the virus causes mucocutaneous lesions after **spreading systemically** through the body
- Viral infections that spread systemically can in turn be divided into:
 - those in which the skin lesions (vesicular) are sites of virus replication and are infectious
 - those in which the skin lesions (maculopapular) are non-infectious and immunologically mediated, although the virus can be shed from other sites

Viral Diseases ...

Virus	Lesion	Virus shedding from lesion
No systemic spread		
Papilloma (wart)	Common wart; plantar wart; genital wart	+
Molluscum contagiosum (poxvirus)	Fleshy papule	+
Orf (poxvirus from sheep, goats)	Papulovesicular	+
Systemic spread		
Herpes simplex, varicella-zoster	Vesicular (neural spread and latency)	+
Coxsackievirus A (9, 16, 23)	Vesicular, in mouth (herpangina)	+
Coxsackievirus A16	Vesicular (hand, foot and mouth disease)	+
Parvovirus B19	Facial maculopapular (erythema infectiosum)	–
Human herpesvirus 6	Exanthem subitum (roseola infantum)	–
Measles	Maculopapular skin rash	–
Rubella, echoviruses (4, 6, 9, 16)	Maculopapular not distinguishable clinically	–
Dengue and other arthropod transmitted viruses	Maculopapular	–

Herpes Infections

- Two viral species of *Simplexvirus* cause herpes: human herpesvirus 1 (HHV-1) and human herpesvirus 2 (HHV-2).
- HHV-2 is the species involved in neonatal herpes
- Herpes simplex virus (HSV) is a medium-sized (120 nm) double-stranded DNA virus of the herpesvirus group
- Human herpesviruses produce various proteins that act as virulence factors:
 - glycoproteins in the viral envelope mediate viral adhesion and fusion
 - other proteins inactivate complement and IgG

Herpes ...

- They cause a wide variety of clinical syndromes, the basic lesion being an intraepithelial vesicle, from which the virus is shed.
- Infection is usually transmitted from the saliva or cold sores of other individuals and frequently by kissing

○Clinical features

- After infection, the virus replicates in cells in the oral mucosa and forms virus-rich vesicles (cold sores and whitlow)
 - mild febrile illness
 - vesicles ulcerate and become coated with a whitish-grey slough
- During the primary infection
 - virus particles enter sensory nerve endings in the lesion
 - transported to the dorsal root (trigeminal) ganglion, where they initiate latent infection in sensory neurones
 - lesion resolves as antibody and CMI responses develop.

Herpes ...

- The latent virus
 - remains in the sensory ganglion for life
 - under certain circumstances can reactivate and spread down sensory nerves to cause cold sores at the site of the original infection



Oral herpes lesions (cold sores)



Whitlow on finger

Herpes ...

○Pathogenesis

- herpesvirus attaches to a host cell (cracks or cuts in mucous membranes) through the fusion of its envelope with the cytoplasmic membrane.
- viral genome is replicated and assembled in the cell's nucleus
- virus acquires an envelope from the nuclear membrane and exits the cell via exocytosis or cell lysis

○Diagnosis

- Clinically by the presence of characteristic, recurring lesions in the genital region and on the lips
- Positive diagnosis is achieved by immunoassay that demonstrates the presence of viral antigen

Papillomavirus infection (Warts)

- Papillomaviruses are 55 nm in diameter, icosahedral, double-stranded DNA viruses and cause skin papillomas (warts)
- > 120 different types of papillomavirus can infect humans and are species specific
 - at least 40 types, including HPV 6, 11, 16 and 18 infect the anogenital tract and other mucosal areas and are sexually transmitted
 - HPV 1 and 4 tend to cause plantar warts
 - HPV 2, 3 and 10 cause warts on the knees and fingers

Papillomavirus ...

○Pathogenesis

- After entering the body via surface abrasions, the virus infects cells in the basal layers of the skin or mucosa. There is no spread to deeper tissues
- The infected cells are stimulated to divide and finally, 1–6 months after initial infection, the mass of infected cells protrudes from the body surface to form a visible papilloma or wart
- Warts can be:
 - filiform with finger-like projections
 - flat topped (plantar warts)
 - a cauliflower-like protuberance (e.g., genital warts)
 - a flat area of dysplasia on the cervix

Papillomavirus ...



Plantar warts on the sole of a foot



Flat warts on the face

Chickenpox and Shingles

- Chickenpox, medically termed **varicella**, is a highly infectious disease caused by
 - **Varicella-zoster virus (VZV)** is a medium-sized (100–200 nm in diameter) double-stranded DNA virus of the herpesvirus group (genus *Varicellovirus*)
- Infection is by
 - inhalation of droplets from respiratory secretions and saliva
 - direct contact from skin lesions
- About 2-3 wks. after infection:
 - patient develops a slight fever and characteristic skin lesions on the back and trunk that spread to the face, neck, and limbs
- Chickenpox lesions begin as **macules**, progress in one to two days to **papules**, and finally become thin-walled, fluid-filled **vesicles** on red bases

Chickenpox ...

- Chickenpox virus can become latent within sensory nerves and may remain dormant for years
- Stress, aging, or immune suppression reactivate the viruses in 15–20% of individuals
- The reactivated viruses:
 - travel down the nerve they inhabit
 - produce an extremely painful skin rash near the distal end of the nerve known as **shingles** or **herpes zoster**

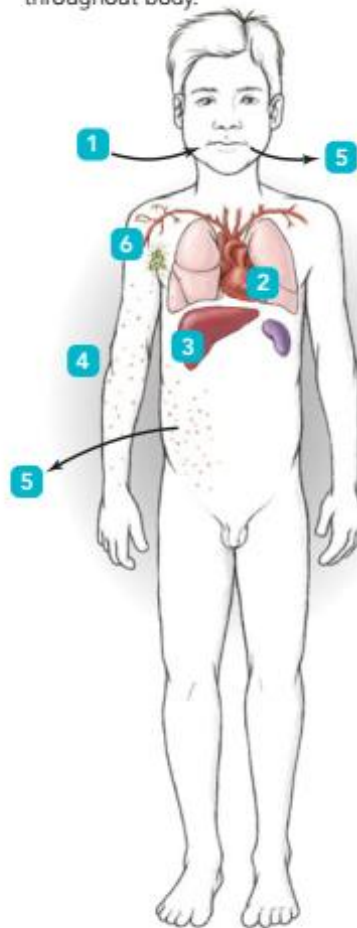
Chickenpox ...

Chickenpox and Shingles

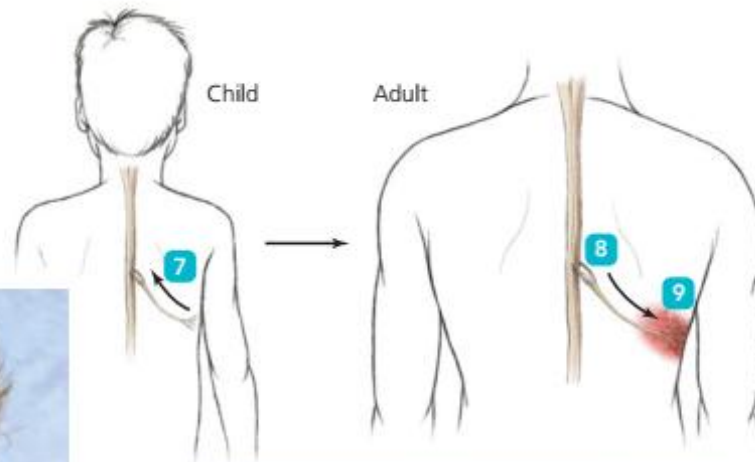
- 1 Inhalation of varicella-zoster virus.
- 2 Infected cells replicate and release virus into the blood (viremia) and lymph.
- 3 Viruses infect liver, spleen, and lymph nodes throughout body.

- 4 Second viremia spreads viruses to skin, causing extensive rash.
- 5 Viruses shed through respiratory droplets and fluid from skin lesions.
- 6 Immune system eliminates viruses from blood and most cells.

- 7 Some viruses become latent (dormant) in nerve ganglia.
- 8 Upon reactivation, viruses travel inside nerve cells down the nerve to infect skin cells.
- 9 Rash develops in the band of skin innervated by the infected nerve.



Characteristic chickenpox lesions



Characteristic shingles rash

Chickenpox ...

○Diagnosis

- characteristic appearance of the lesions
- molecular tests to detect VZV DNA in swabs from the vesicles
- immunofluorescence tests on skin lesion scrapings using VZV-specific monoclonal antibodies
- isolating VZV in cell culture

Measles (Rubeola)

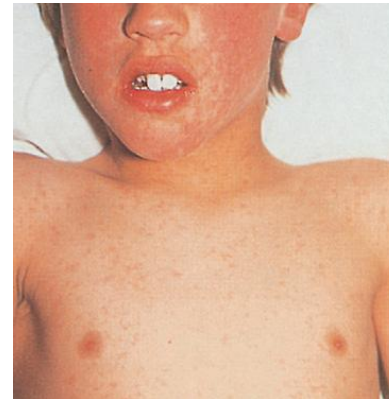
- Measles virus (*Morbillivirus*) is a single-stranded RNA virus with an enveloped helical capsid.

○ Signs and Symptoms

- include fever, sore throat, headache, dry cough, and conjunctivitis
- after two days of illness, lesions called **Koplik's spots** appear on the mucous membrane of the mouth
- Red maculopapular lesions appear on the head and spread over the body



Koplik's spots - minute white dots on the inflamed buccal mucosa of a patient with measles



Maculopapular rash on the face and trunk of a patient with measles

Measles ...

○Pathogen and Virulence Factors

- adhesion proteins
- envelope has a fusion protein:
 - triggers fusion between an infected cell and its neighbors
 - allows the virus to pass from cell to cell and avoid antibodies

Measles ...

○Pathogenesis

- Measles virus infects cells of the respiratory tract before spreading via lymph and blood throughout the body to infect
 - the conjunctiva
 - urinary tract
 - small blood vessels
 - lymphatics
 - CNS
 - skin
- Cytotoxic T cells kill infected cells, causing most of the symptoms
- Most patients recover within 2-3 wks.
- 1-5% of infected children die

Measles ...

○Diagnosis

- signs of measles, particularly Koplik's spots, are sufficient for diagnosis
- serological testing confirms the presence of measles antigen in respiratory and blood specimens

Other Viral Rashes

○Erythema Infectiosum

- a respiratory disease caused by a single-stranded DNA (ssDNA) virus called B19 virus in the genus *Erythrovirus*, family *Parvoviridae*
- rash begins on the cheeks and spreads to the arms, thighs, buttocks, and trunk
 - facial rash appears as if the patient has been slapped



Other Viral ...

○Roseola

- characterized by an abrupt fever, sore throat, enlarged lymph nodes, and a faint pink rash on the face, neck, trunk, and thighs
- A herpesvirus in the genus *Roseolovirus*, known as *human herpesvirus 6* (HHV-6) causes roseola

○Coxsackievirus Infection

- several serotypes of coxsackie A virus cause lesions of the mouth and pharynx called **herpangina** (lesions resemble those of herpes)
- Another serotype of coxsackie A virus causes hand-foot-and-mouth disease involving lesions on the extremities and in the mouth

Mycoses of the Skin, Soft tissues, Hair and Nails

Superficial Mycoses

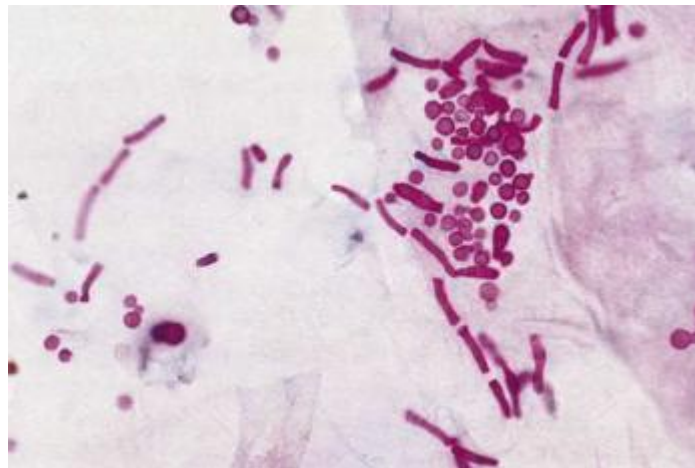
- Agents of superficial mycoses are fungi that colonize the keratinized outer layers of the skin, hair, and nails
- Infections caused by these organisms elicit little or no host immune response and are non-destructive and thus asymptomatic.
- They are usually of cosmetic concern only and are easy to diagnose and treat

Pityriasis (tinea) versicolor

- Pityriasis versicolor is a common superficial fungal infection that is seen worldwide.
- In tropical environments, it may affect 30% to 35% of the population
- It is caused by the lipophilic yeast species of the *Malassezia furfur* complex: *M. furfur*, *M. sympodialis*, *M. globosa*, *M. restricta*, *M. slooffiae*, *M. obtusa*, *M. dermatis*, *M. japonica*, and *M. yamatoensis*
- Human infection results from the direct or indirect transfer of infected keratinous material from one person to another

Pityriasis ...

- When viewed in skin scrapings, members of the *M. furfur* complex appear as:
 - clusters of spherical or oval, thick-walled yeast-like cells, 3 to 8 μm in diameter
 - The yeast cells may be mixed with short, infrequently branched hyphae that tend to orient end to end



Pityriasis ...

□ Clinical Syndromes

- The lesions of pityriasis versicolor are small hypopigmented macules
- The upper trunk, arms, chest, shoulders, face, and neck are most often involved, but any part of the body may be affected
- The lesions are irregular, well-demarcated patches of discoloration that may be raised and covered by a fine scale.
- Because species of the *M. furfur* complex tend to interfere with melanin production, lesions are
 - hypopigmented in dark-skinned individuals
 - pink to pale brown in light-skinned individuals

Pityriasis ...



Pityriasis versicolor. Multiple, pale brown,
Hyperpigmented patches on chest and shoulders

Pityriasis ...

□ Laboratory Diagnosis

- made by the direct visualization of the fungal elements on microscopic examination of epidermal scales in 10% KOH with or without calcofluor white.

Tinea nigra

- Tinea nigra is a superficial fungal infection caused by the black fungus *Hortaea werneckii* (formerly *Exophiala werneckii*)
- Microscopically, *H. werneckii* appears as
 - dematiaceous (melanin [brown to black] pigmented), frequently branched, septate hyphae, 1.5 to 3.0 μm wide
 - arthroconidia and elongate budding cells are also present
- It is contracted by traumatic inoculation of the fungus into the superficial layers of the epidermis
- Tinea nigra appears as
 - a solitary, irregular, pigmented (brown to black) macule, usually on the palms or soles
 - there is no scaling or invasion of hair follicles, and the infection is not contagious

Tinea nigra ...



Tinea nigra ...

- **Laboratory Diagnosis**

- Tinea nigra is easily diagnosed by microscopic examination of skin scrapings placed in 10% - 20% KOH.
- The pigmented hyphae and yeast forms are confined to the outer layers of the stratum corneum and are easily detected on H&E

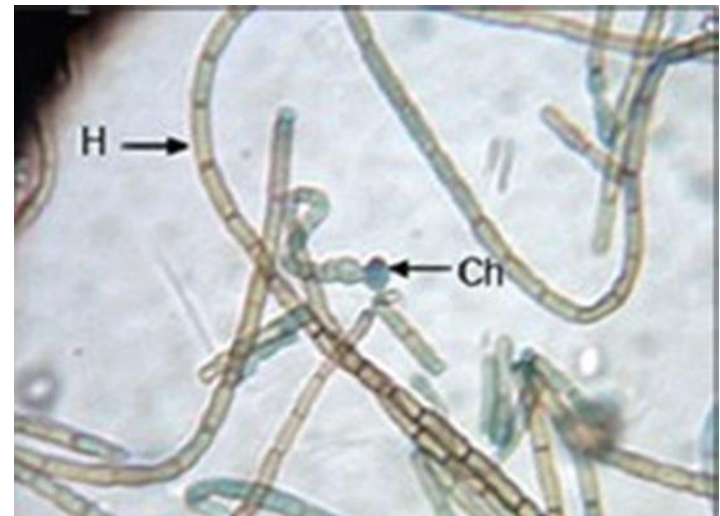


Black piedra

- Fungal infection of the hair, primarily the scalp
- The causative agent of black piedra is *Piedraia hortae*
- The organism grows as pigmented (brown to reddish-black) mold
- Black piedra
 - presents as small, dark nodules that surround the hair shafts
 - is asymptomatic and generally involves the scalp
 - the hyphal mass is held together by a cement-like substance and contains asci and ascospores, which is the sexual phase of the fungus.

Black piedra ...

- Examination of the nodule reveals branched, pigmented, hyphae held together by a cement-like substance



White piedra

- White piedra is a superficial infection of hair caused by yeastlike fungi of the genus *Trichosporon*
 - *T. ovoides* (causes scalp hair white piedra)
 - *T. inkin* (causes most cases of pubic white piedra)
 - *T. asahii*
- Microscopic examination reveals
 - hyphal elements, arthroconidia (rectangular cells resulting from the fragmentation of hyphal cells)
 - blastoconidia (budding yeast cells)

White piedra ...



Left: Creamy and soft colonies of *T. beigelii* Right: hyphae and barrel-shaped arthroconidia of *Trichosporon beigelii*

Cutaneous mycoses/Dermatophytoses

- a complex of diseases caused by any of several species of taxonomically related filamentous fungi in the genera *Trichophyton*, *Epidermophyton*, and *Microsporum*, known collectively as **the dermatophytes**
- All have in common
 - the ability to invade the skin, hair, or nails
 - are keratinophilic and keratinolytic; thus they are able to break down the keratin surfaces of these structures.
- In the case of skin infections, the dermatophytes invade only the upper, outermost layer of the epidermis, which is the stratum corneum. Penetration below the granular layer of the epidermis is rare
- With hair and nails, being part of the skin, only the keratinized layers are invaded.

Cutaneous mycoses ...

- The various forms of dermatophytosis are referred to as “tineas” or ringworm
- Clinically, the tineas are classified according to the anatomic site or structure affected
 - **tinea capitis** of the scalp, eyebrows, and eyelashes
 - **tinea barbae** of the beard
 - **tinea corporis** of the smooth or glabrous skin
 - **tinea cruris** of the groin
 - **tinea pedis** of the foot
 - **tinea unguium** of the nails (also known as **onychomycosis**)

Cutaneous mycoses ...

- Dermatophytes can be classified into three different categories based on their natural habitat
 - geophilic
 - zoophilic
 - anthropophilic

Cutaneous mycoses ...

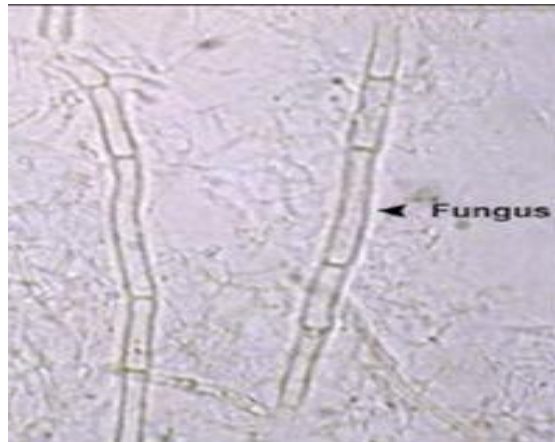
- The classic pattern of dermatophytosis is the “ringworm” pattern of a ring of inflammatory scaling with diminution of inflammation toward the center of the lesion.
- Tineas of hair-bearing areas often present as raised, circular or ring-shaped patches of alopecia with erythema and scaling or as more diffusely scattered papules, pustules, vesicles, and kerions (severe inflammation involving the hair shaft)



Cutaneous mycoses ...

○Laboratory Diagnosis

- relies on the demonstration of fungal hyphae by direct microscopy of skin, hair, or nail samples
- filamentous, hyaline hyphal elements characteristic of dermatophytes may be seen in skin scrapings, nail scrapings, and hairs.



Cutaneous and mucocutaneous candidiasis

- *Candida* species are important and common human pathogens causing a wide spectrum of superficial, mucocutaneous and systemic/deep disease
 - Superficial infections are frequently community acquired and responsible for considerable morbidity.
 - In contrast, deep seated, invasive, and systemic *Candida* infections are usually nosocomial in origin.

Candidiasis ...

- *Candida* are small (4–6 μ m), oval, thin-walled yeast-like fungi that reproduce by budding or fission.
- On culture media, *Candida* species form smooth, creamy white, glistening colonies.
- The genus *Candida* is comprised of over 200 species and constitutes an extremely diverse yeast species whose common bond is the absence of a sexual cycle

Candidiasis ...

- > 17 different species of *Candida* reported as pathogens
 - more than 90% of invasive infections are attributed to five species; *Candida albicans*, *Candida glabrata*, *Candida parapsilosis*, *Candida tropicalis*, and *Candida krusei*
- *Candida albicans* remains the major fungal pathogen of humans and the most common cause of mucosal and systemic fungal infection

Candidiasis ...

- Species of *Candida*, including *C. albicans* are part of the normal endogenous microbial flora
 - candidiasis is usually the result of autoinfection during a metabolic or an immunologic disturbance.
 - the organisms may be recovered from the oropharynx GI, genitourinary tract, and skin.

Pathogenesis and Clinical Pictures

A. Cutaneous Candidiasis (Candidosis, Moniliasis)

- Superficial infection occurring on moist cutaneous sites

Epidemiology and Etiology

- **Age:** any age; Infants (diaper area, mouth).
- **Sex:** both; Women (vulvovaginitis), men (balanitis)
- **Etiology:** *Candida albicans*; uncommonly, other *Candida* species.
- **Ecology:** *C. albicans* is seldom recovered from skin of normal individuals; usually endogenous infection. *Candida* is a normal inhabitant of mucosal surfaces of oropharynx and GI tract. In males with balanitis, *Candida* may be transmitted from female sexual partner.

Cutaneous...

- Occupation: Persons who immerse their hands in water: housewives, mothers of young children, health care workers, bartenders, florists
- Other predisposing factors: diabetes, obesity, hyperhidrosis, heat, maceration, systemic and topical corticosteroids, chronic debilitation.

Cutaneous...

○ **Candidiasis in occluded skin** - Occurs where occlusion and maceration create warm, moist microecology.

- Occlusive dressing, under cast, back in hospitalized patient
- Folliculitis usually at sites of occlusion
- Diaper dermatitis may be primary or secondary infection

○ **Intertrigo**

- Body folds: axillae, inframammary, groins, intergluteal
- Webspaces: fingers (**erosio interdigitalis blastomycetica**); associated with frequent wetting of hands, toes.

○ **Genital candidiasis**

- Balanitis
- Vulvitis

Cutaneous...

- **Candida Paronychia and Onychia** associated with frequent wetting of hands

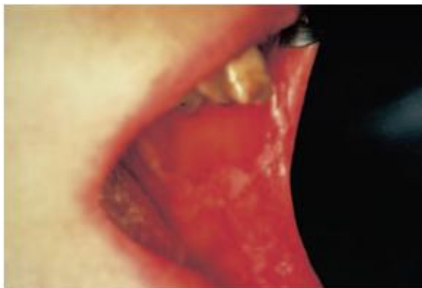


Clinical ...

B. Mucosal Candidiasis

○ Superficial mucosal candidiasis

- Vaginal and Vulvovaginal Candidiasis
- Oropharyngeal Candidiasis
 - Erythematous (atrophic) candidiasis
 - Pseudomembranous candidiasis (thrush)
 - Candida leukoplakia (hyperplastic candidiasis)
 - Angular cheilitis



Pseudomembranous oral candidiasis.



Angular cheilitis.



Laboratory Diagnosis

- Superficial or mucocutaneous candidiasis is diagnosed by finding the fungus in tissue scraping and culture.

Definitive diagnosis is made by the histopathologic demonstration of the invasion of tissue by the yeast.

- Specimens from surface lesions, mouth, vaginal, sputum, exudates etc., are examined using different methods

Lab Dx cont'd

○Direct examination

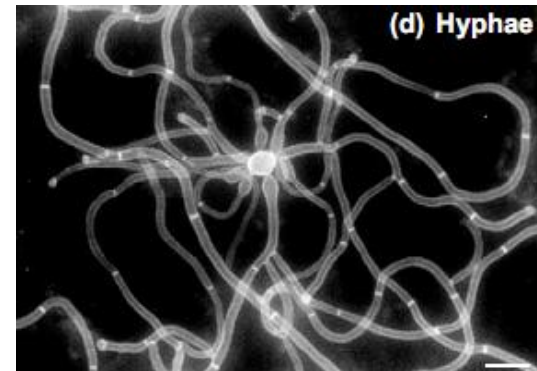
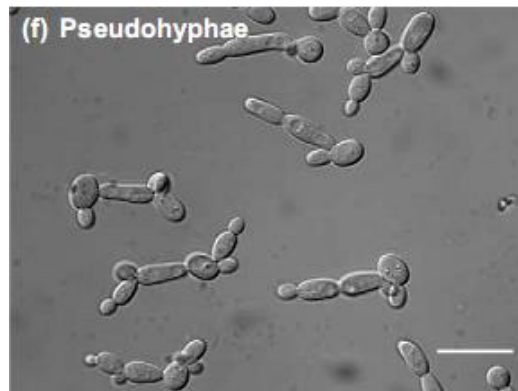
•KOH

- Exposed lesions can usually be easily diagnosed by clinical appearance together with finding typical budding yeast cells and pseudohyphae and/or true hyphae in lesion scrapings treated with KOH.

•Gram- stain

- Gram stain smears show large gram positive budding yeast cells with pseudohyphae.

Lab Dx cont'd



Subcutaneous Mycoses

- are clinical syndromes caused by multiple fungal etiologies
- the causative agents of subcutaneous mycoses
 - have low pathogenic potential
 - are commonly isolated from soil, wood, or decaying vegetation
 - exposure is largely occupational or related to hobbies (e.g., gardening, wood gathering)
 - infected patients generally have no underlying immune defect

Common Agents of Subcutaneous Mycoses

Disease	Etiologic Agent(s)	Typical Morphology in Tissue	Usual Host Reaction
Sporotrichosis	<i>Sporothrix schenckii</i> , <i>S. brasiliensis</i> , <i>S. globosa</i> , and <i>S. luriei</i>	Pleomorphic, spheric to oval or cigar-shaped yeasts, 2- to 10- μ m diameter with single or multiple (rare) buds See Fig. 63.3	Mixed suppurative and granulomatous Splendore-Hoepli material surrounds fungus (asteroid body) See Fig. 63.4
Chromoblastomycosis	<i>Cladophialophora</i> (<i>Cladosporium</i>) <i>carrionii</i> <i>Fonsecaea pedrosoi</i> <i>Phialophora verrucosa</i> <i>Rhinocladiella</i> spp. <i>Exophiala</i> spp.	Large, 6- to 12- μ m diameter, spheric, thick-walled, brown muriform cells (sclerotic bodies) with septations along one or two planes; pigmented hyphae may be present See Fig. 63.6	Mixed suppurative and granulomatous Pseudoepitheliomatous hyperplasia
Eumycotic mycetoma	<i>Phaeoacremonium</i> spp. <i>Fusarium</i> spp. <i>Aspergillus nidulans</i> <i>Scedosporium</i> spp. <i>Madurella</i> spp. <i>Exophiala jeanselmei</i> , among others	Granules, 0.2 to several millimeter diameter, composed of broad (2- to 6- μ m), hyaline (pale granules), or dematiaceous (black granules) septate hyphae that branch and form chlamydoconidia	Suppurative with multiple abscesses, fibrosis, and sinus tracts; Splendore-Hoepli material
Subcutaneous entomophthoromycosis	<i>Basidiobolus ranarum</i> <i>Conidiobolus coronatus</i>	Short, poorly stained hyphal fragments, 6- to 25- μ m diameter, nonparallel sides, pauciseptate, random branches See Fig. 63.10	Eosinophilic abscesses and granulation tissue, Splendore-Hoepli material around hyphae
Subcutaneous phaeohyphomycosis	<i>Exophiala jeanselmei</i> <i>E. dermatitidis</i> <i>Alternaria</i> spp. <i>Chaetomium</i> spp. <i>Curvularia</i> spp. <i>Phialophora</i> spp., among others	Pigmented (brown) hyphae, 2- to 6- μ m diameter, branched or unbranched, often constricted at prominent septations, yeast forms and chlamydoconidia may be present See Fig. 63.11	Subcutaneous cystic or solid granulomas; overlying epidermis rarely affected

Subcutaneous ...



Papules and draining sinuses of mycetoma on the foot

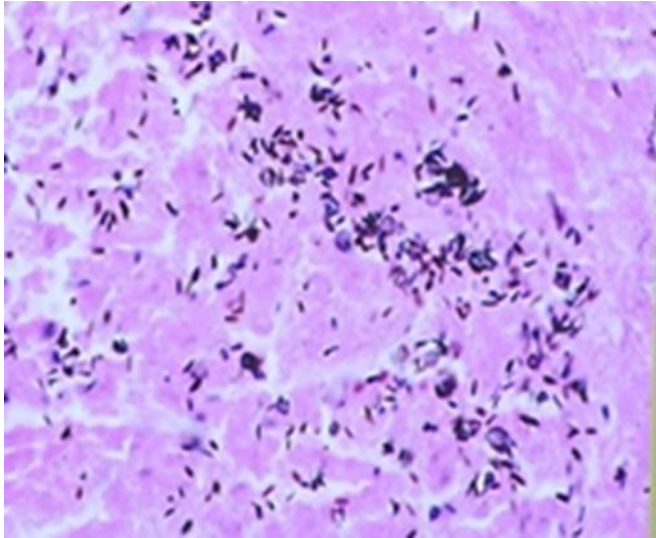
(a) Characteristic lesions of lymphocutaneous sporotrichosis of the forearm (b) Primary lesion, which has ulcerated, in the same patient

Subcutaneous ...

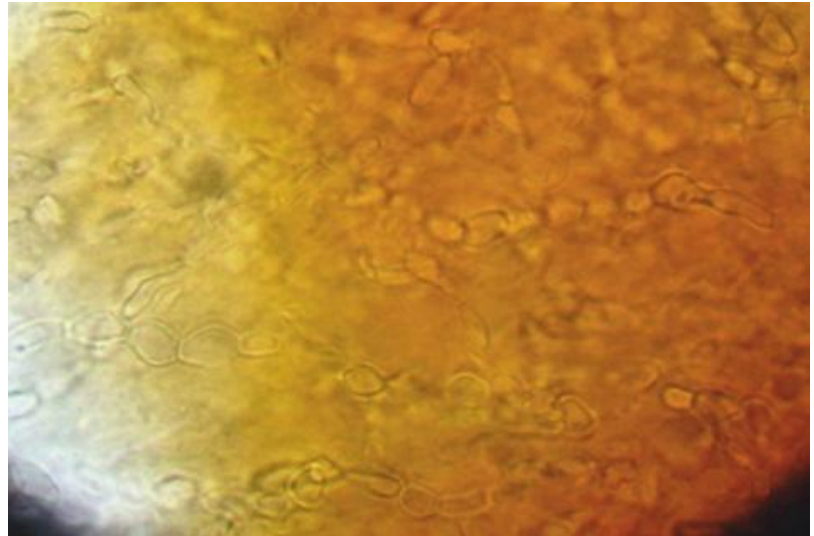


Nodular, cauliflower-like
lesions of chromoblastomycosis

Subcutaneous ...



PAS stained tissue section showing cigar-shaped budding yeast-like cells of *S. schenckii*.



Intertwined, broad hyphae of *M. mycetomatis*

II. Microbial Diseases of Bones and Joints

- **Osteomyelitis:** infection of bone and bone marrow, particularly the metaphysis of long bones. Infection comes:
 - directly from a focus of infection (e.g., a compound fracture)
 - from circulating infection
 - classified as:
 - **Acute**
 - acute haematogenous osteomyelitis
 - acute contiguous-focus osteomyelitis
 - acute ischaemic-neuropathic osteomyelitis

Bones and Joints ... Osteomyelitis ...

- **Subacute**

- device-related osteomyelitis
- vertebral body osteomyelitis
- fungal osteomyelitis (rare)

- **Chronic**

- pyogenic ('bacterial') osteomyelitis
- actinomycosis (rare)
- brucellosis (rare)
- tuberculous osteomyelitis

Osteomyelitis: Type, Patient Profile, and Common Causes

Type of Osteomyelitis/Patient Profile	Common Causes
Hematogenous	Usually only one organism
Infants (<1 year)	<i>Staphylococcus aureus</i> , <i>Streptococcus agalactiae</i> (group B <i>Streptococcus</i>), <i>Escherichia coli</i>
Children (1–16 years)	<i>S aureus</i> , <i>Streptococcus pyogenes</i> (group A <i>Streptococcus</i>), <i>Haemophilus influenzae</i>
Persons >16 years	<i>S aureus</i> , coagulase-negative staphylococci (e.g., <i>Staphylococcus epidermidis</i>), gram-negative rod-shaped bacteria (e.g., <i>E coli</i> , <i>Pseudomonas</i> , <i>Serratia</i>)
Contiguous spread*	More likely to be polymicrobial
Microbiology depends on the primary site of infection	<i>S aureus</i> , <i>S pyogenes</i> (group A <i>Streptococcus</i>), <i>Enterococcus</i> (i.e., group D; <i>E faecalis</i> and <i>E faecium</i>), coagulase-negative staphylococci (e.g., <i>Staphylococcus epidermidis</i>), gram-negative rod-shaped bacteria (e.g., <i>Escherichia coli</i> , <i>Pseudomonas</i> , <i>Serratia</i>), anaerobes (e.g., <i>Prevotella</i> , <i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i>)
Diabetic foot	<i>S aureus</i> , <i>Streptococcus</i> , <i>Enterococcus</i> , gram-negative rod-shaped bacteria (e.g., <i>Proteus mirabilis</i> , <i>Pseudomonas</i>), anaerobes (e.g., <i>Prevotella</i> , <i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i>)

Bones and Joints ...

- **Infective arthritis:** painful erythematous joint with collection of purulent material in the joint space
 - it is classified clinically into:
 - **acute**, usually from pyogenic bacteria ('septic arthritis'), or viruses
 - **chronic**, usually tuberculous, spirochaetal or fungal
 - the *route* is usually blood spread from a primary focus; occasionally caused directly by trauma, surgery or animal or human bites

Synovial Fluid Analysis to Differentiate Septic Arthritis from Noninflammatory and Inflammatory Arthritis

Laboratory Test	Normal Synovial Fluid	Septic Arthritis	Noninflammatory Arthritis	Inflammatory Arthritis
Clarity and color	Clear	Opaque, yellow to green	Clear, yellow	Translucent, yellow, or opalescent
Viscosity	High	Variable	High	Low
White blood cells/mm ³	<200	>100,000*	200–2000	2000–10,000
Percent polymorphonuclear leukocytes	<25%	>75%	<25%	≥50%
Total protein g/dL	1–2	3–5	1–3	3–5
Glucose concentration (concentration in joint relative to concentration in blood)	Nearly equal	<25%	Nearly equal	50–80%
Culture	Negative	Often positive in nongonococcal arthritis; usually negative in gonococcal arthritis	Negative	Negative
Disease or condition	Normal	Septic arthritis	Osteoarthritis, trauma to joint	Rheumatoid arthritis Reiter disease Partially treated septic joint Fungal or viral infection Gout or pseudogout Acute rheumatic fever